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2027 CYCLE

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CUET UG

Chemistry

Biomolecules

Official Syllabus Unit X · Class 12 · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus 2026, the current 2027 baseline, Unit X)	Skills the questions test
Carbohydrates	Classification of carbohydrates, Methods of preparation of glucose, Structure of Glucose, D&L configuration of monosaccharides, method of preparation of fructose, structure of fructose, disaccharides: sucrose, maltose and lactose, polysaccharides: starch, cellulose and glycogen, importance of carbohydrates	Classifying sugars as mono/di/poly and reducing/non-reducing, reciting the glucose evidence chain (open chain → ring), naming the glycosidic links, and matching starch/cellulose/glycogen to their roles.
Amino acids and proteins	Amino acids and their classifications, structure and types of proteins, denaturation of proteins	Sorting essential vs non-essential amino acids, ordering the four protein structures, and explaining what denaturation does and does not break.
Enzymes and vitamins	Enzymes, mechanism of enzyme action (elementary idea), Vitamins: Types, sources and deficiency diseases	Describing lock-and-key specificity, pairing each vitamin with its deficiency (A-night blindness, B_1-beriberi, C-scurvy, D-rickets, B_{12}-pernicious anaemia) and its fat/water solubility.
Nucleic acids and hormones	Nucleic acids: types, composition, structure and biological importance, Hormones: classification, biological importance and deficiency diseases	Contrasting DNA vs RNA (sugar, bases, strands), naming the nucleotide parts, and matching hormones to their classes and deficiency diseases.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Chemistry paper maps strictly to NCERT Class 12, and Unit X closes the syllabus with the memory unit — facts, names and pairings rewarded far more than calculation.

Carbohydrates: The Fuel Family

Bread, rice, potato and sugar share one chemistry: **carbohydrates** — polyhydroxy aldehydes or ketones (or things that hydrolyse into them). They are the body's instant fuel, plants' structural skeleton, and the exam's favourite classification exercise.

- **Monosaccharides:** cannot be hydrolysed further — glucose and fructose (6 C, **hexoses**), ribose (5 C, **pentose**). Free aldehyde = **aldose**; ketone = **ketose**.
- **Reducing vs non-reducing:** free aldehyde/ketone group → reduces Tollen's/Fehling's → **reducing sugar** (all monosaccharides, maltose, lactose); **sucrose alone fails** — its two anomeric carbons are busy in the glycosidic bond.
- **Disaccharides:** two monosaccharides joined by a **glycosidic bond** — sucrose = glucose + fructose (α-1,2), maltose = glucose + glucose (α-1,4), lactose = galactose + glucose (β-1,4).
- **Polysaccharides:** starch (storage, amylose helix + amylopectin branches), cellulose (β-1,4 straight chains — wood, cotton, paper; humans cannot digest it), glycogen (animal starch, heavily branched, liver and muscles).

Glucose: preparation and the evidence chain

- **Two preparations:** hydrolyse cane sugar (sucrose) with dilute acid → glucose + fructose (ethanol precipitates the glucose); or hydrolyse starch with dilute H_2SO_4 at 393 K (the industrial route).
- **Structure evidence, step by step:** (1) formula $C_6H_{12}O_6$; (2) reduces Tollen's/Fehling's → a **free aldehyde**; (3) forms a penta-acetate with acetic anhydride → **five OH groups**; (4) does NOT give the 2,4-DNP adduct or bisulphite adduct strongly, and does not restore pink colour to Schiff's reagent fast → the CHO is **not free in solution**; (5) two stereoisomeric glucosides form with methanol/HCl → a **hemiacetal** exists → the molecule is a **ring**; (6) n-hexane product on HI reduction → straight 6-carbon chain.
- **The pyranose ring:** C-5′s OH attacks C-1′s aldehyde → a 6-membered (5 C + 1 O) **pyranose** ring with a new **anomeric** carbon (C-1). α-glucose: the anomeric OH is trans to CH_2OH; β-glucose: cis.
- **D and L configuration:** the **penultimate** carbon's OH decides — right side = D (like D-glucose), left = L. D/L says NOTHING about optical rotation sign (+/− is measured, not predicted — the classic trap). Natural glucose is D-(+)-glucose.
- **Fructose:** the ketohexose twin — same formula, C=O at C-2; from the same sucrose hydrolysis; furanose ring form; also a reducing sugar (tautomerises to an aldose under test conditions).

Amino Acids and Proteins: The Workforce

- **Amino acid structure:** $H_2N-CH(R)-COOH$ — amine and acid on one carbon. In water it exists as a **zwitterion** ($H_3N^+-CHR-COO^-$) — amphoteric, high melting point, water-soluble.
- **Classification: essential** amino acids must come from diet (valine, leucine, isoleucine, lysine, methionine, phenylalanine, threonine, tryptophan… — ten in all); **non-essential** ones the body synthesises (glycine, alanine, aspartic acid…).
- **Protein structures, four floors: primary** (the amino-acid sequence, peptide bonds); **secondary** (local folding into α-helix or β-sheet, held by H-bonds); **tertiary** (the whole 3-D fold — S–S bridges, ionic and van der Waals forces); **quaternary** (several chains assembled, e.g. haemoglobin's four subunits).
- **Denaturation:** heat, acid, base or agitation wreck the secondary/tertiary folding — the protein uncoils, globular turns fibrous, **biological activity is lost** (boiled egg). The **primary sequence survives** — denaturation never breaks peptide bonds. (Renaturation sometimes restores activity.)
- **Peptide bond:** $-CO-NH-$ between amino acids; two make a dipeptide, many make a polypeptide — and insulin is a 51-residue two-chain protein.

Enzymes and Vitamins: Regulation and Defence

- **Enzymes:** globular proteins that catalyse reactions at body temperature with astonishing **specificity**. The **lock-and-key** model: the substrate fits the enzyme's active site like a key; modern **induced-fit** refinement lets the site flex. Names: substrate + "-ase" (urease splits urea, invertase splits sucrose, zymase ferments glucose).
- **Vitamins — fat-soluble (store in liver): A** (night blindness), **D** (rickets — bone softening), **E**, **K** (poor clotting).
- **Vitamins — water-soluble (not stored, daily need): B_1 thiamine** (beriberi), **B_2**, **B_6**, **B_{12} cyanocobalamin** (pernicious anaemia), **C ascorbic acid** (scurvy). The pairing drill — vitamin ↔ deficiency ↔ solubility — is guaranteed marks.

Nucleic Acids and Hormones: Memory and Messengers

- **Nucleotide = base + sugar + phosphate**; the sugar-phosphate ester chain forms the backbone.
- **DNA:** sugar = **β-D-2-deoxyribose** (no OH on C-2); bases = A, T, G, C; **double helix** (Watson-Crick) with A=T (2 H-bonds) and G≡C (3 H-bonds) pairing; carries heredity — replication hands the blueprint down.
- **RNA:** sugar = **ribose**; bases = A, U, G, C (uracil replaces thymine); usually **single-stranded**; runs protein synthesis — mRNA reads, tRNA carries, rRNA builds.
- **Hormones:** chemical messengers. **Steroids** (testosterone, cortisol), **amines** (adrenaline, thyroxine), **peptides/proteins** (insulin, growth hormone). Deficiency pairs: insulin → diabetes mellitus; thyroxine (iodine) → goitre; growth hormone (childhood) → dwarfism, (excess) → gigantism; ADH → diabetes insipidus.

The Whole Chapter in One Picture

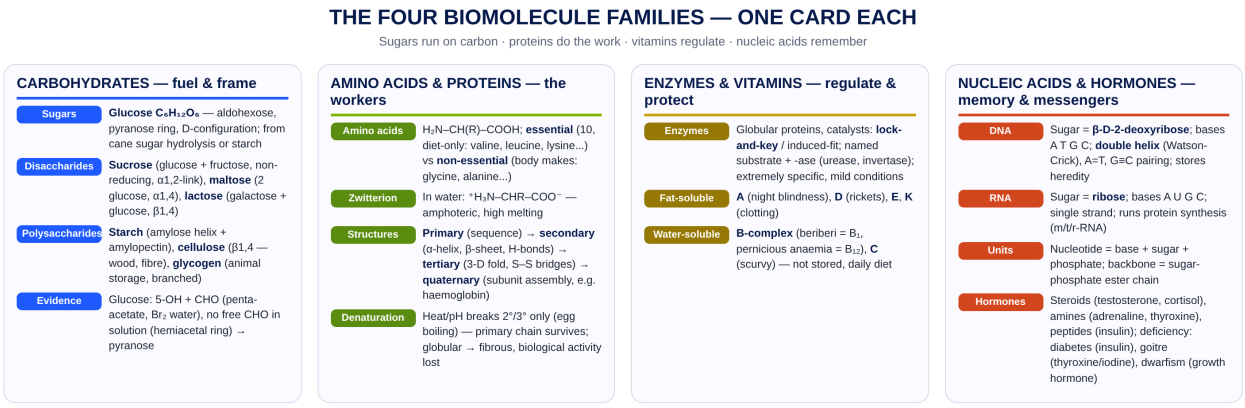


Figure 1: The four biomolecule families as chapter map: carbohydrates with glucose's evidence chain and the sugar links, amino acids and the four protein floors with denaturation, enzymes and the vitamin-deficiency pairs, and nucleic acids with DNA-vs-RNA contrasts plus hormone classes.

Practice Questions (Exam Pattern)

Ten questions in the CUET UG style. Attempt all of them — the paper gives +5 for each correct answer and takes nothing away for a wrong one.

Q1. Which sugar is NOT a reducing sugar?

(A) glucose (B) maltose
(C) lactose (D) sucrose

Q2. The glycosidic linkage in sucrose connects:

(A) glucose C1 and glucose C4 (B) glucose C1 and fructose C2
(C) galactose and glucose (D) two fructose units

Q3. Cellulose is a polymer of β-glucose joined by:

(A) α-1,4 linkages (B) β-1,4 linkages
(C) α-1,6 linkages (D) β-1,2 linkages

Q4. The D-configuration of glucose refers to:

(A) its rotation of light to the right (B) the OH on the penultimate carbon being on the right
(C) its D-shaped ring (D) its density

Q5. Glucose's failure to give an instant Schiff's test shows that:

(A) it is not an aldehyde at all (B) most molecules exist as the hemiacetal ring, with little free CHO
(C) it is a ketose (D) it lacks OH groups

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Q6. Which amino acid is essential in the human diet?

(A) glycine **(B)** alanine
(C) lysine **(D)** aspartic acid

Q7. Denaturation of a protein destroys:

(A) the primary structure **(B)** the peptide bonds
(C) the secondary and tertiary structure **(D)** all covalent bonds

Q8. The base NOT present in DNA is:

(A) adenine **(B)** guanine
(C) uracil **(D)** cytosine

Q9. Vitamin deficiency matching — which pair is correct?

(A) Vitamin C — beriberi **(B)** Vitamin B_1 — scurvy
(C) Vitamin D — rickets **(D)** Vitamin K — night blindness

Q10. Insulin deficiency causes:

(A) goitre **(B)** rickets
(C) diabetes mellitus **(D)** pernicious anaemia

Detailed Solutions

Sol. Q1. Reducing sugars carry a free anomeric carbon. Sucrose's glycosidic bond ties up BOTH anomeric carbons (glucose C1 to fructose C2) — no free CHO/C=O, no reduction. Every other option reduces Tollen's/Fehling's.

Answer: (D)

Sol. Q2. Sucrose links the anomeric carbons of its two parents: glucose C1 (α) to fructose C2 (β) — the α-1,2 glycosidic bond. Maltose is α-1,4 of two glucoses; lactose joins galactose to glucose.

Answer: (B)

Sol. Q3. Cellulose's β-1,4 links build straight, flat chains that H-bond into fibres — the reason paper and cotton hold shape, and why human enzymes cannot digest them. Starch uses α-links we CAN digest.

Answer: (B)

Sol. Q4. D/L reads the penultimate carbon's OH in the Fischer projection — right = D. Optical rotation is an independent measurement: D-glucose is (+) but D-fructose is (−). Configuration ≠ rotation sign.

Answer: (B)

Sol. Q5. The open-chain aldehyde exists only fleetingly — nearly all glucose sits as the pyranose hemiacetal ring. Schiff's reagent (and 2,4-DNP/bisulphite) reacts too slowly to catch it.

Answer: (B)

Sol. Q6. The ten essential amino acids must come from food — lysine is the classic example. Glycine, alanine and aspartic acid are non-essential: the body builds them itself.

Answer: (C)

Sol. Q7. Denaturation unwinds the H-bonded helices and 3-D folds (secondary/tertiary) but never touches the covalent peptide-bonded sequence. Boiled egg: unfolded, still a chain of amino acids.

Answer: (C)

Sol. Q8. DNA's alphabet is A, T, G, C — uracil belongs to RNA, replacing thymine (with ribose as the sugar instead of deoxyribose).

Answer: (C)

Sol. Q9. Vitamin D builds bones — its shortage bends them (rickets). The mislabelled pairs: beriberi is B_1, scurvy is C, night blindness is A.

Answer: (C)

Sol. Q10. Insulin manages blood glucose; its shortage is diabetes mellitus. Goitre tracks thyroxine/iodine; pernicious anaemia tracks B_{12}.

Answer: (C)

Wrong-Answer Clinic — the one-line corrections

Dropped marks above? Find your slip here and rewrite the corrected rule in your own words.

If you answered...	Your slip was...	The corrected rule, in one line
Q1 option A	Calling every sugar reducing	Only sucrose locks both anomeric carbons — the lone non-reducing sugar on the list
Q4 option A	Mixing D/L with +/−	D/L is configuration (Fischer OH side); +/− is measured rotation — unrelated signs

If you answered...	Your slip was...	The corrected rule, in one line
Q6 option B	Trusting familiar names	Essential = cannot be synthesised — memorise the ten, lysine among them
Q7 option A	Overestimating the damage	Denaturation spares the primary sequence — only H-bonded folding collapses
Q10 option A	Cross-wiring glands	Match messenger to gland: insulin-pancreas-glucose, thyroxine-thyroid-iodine

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